

Developer Mock Test Results

Student: Garre prahas

Name:	Garre prahas	Student ID:	116
Date:	2026-03-05 12:44:16	Performance:	■■ Needs Improvement
Overall Score:	7.4/25 (29.6%)	Test ID:	1916d658-25c0-419a...
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	2/10	20.0%	■■ Needs Work
■ MCQ/Theory	4/10	40.0%	■■ Needs Work
■ Coding	1.4/5	28.0%	■■ Needs Work

Detailed Question Review

■ APTITUDE SECTION (2/10 - 20.0%)

■ Q1. The average of 3 numbers is 15. What is the sum of the 3 numbers?

Your Answer: 40

Correct Answer: 45

■ The correct answer is 45 because the average of 3 numbers is calculated by dividing the sum of the numbers by 3. To find the sum, we can multiply the average by 3: $15 * 3 = 45$. The user's answer of 40 is likely incorrect because they may have misunderstood the calculation or forgotten to multiply the average by the total number of values.

■ Q2. If 12 workers can build a house in 8 days, how many workers will be needed to build the same house in 6 days?

Your Answer: 16 workers

■ Correct! Good understanding.

■ Q3. What is the next number: 1, 3, 6, 10, 15, ?

Your Answer: 20

Correct Answer: 21

■ The correct answer is 21 because the pattern of the sequence is obtained by adding 2, 3, 4, 5, and so on to the previous term. The key calculation step is: $1 + 2 = 3$, $3 + 3 = 6$, $6 + 4 = 10$, $10 + 5 = 15$, $15 + 6 = 21$. The user's likely mistake was not recognizing the increasing increment in the sequence, which led them to choose 20 instead of the correct answer.

■ Q4. What is the next number: 1, 2, 4, 8, 16, ?

Your Answer: 24

Correct Answer: 32

■ The correct answer is 32 because the pattern of the sequence is obtained by doubling the previous number. The key calculation step is: $16 * 2 = 32$. The user's answer, 24, is likely incorrect because it breaks the doubling pattern, suggesting a misunderstanding of the sequence's progression.

■ **Q5. A product is on sale for 20% off its original price of \$80. How much will you pay for the product?**

Your Answer: \$65

Correct Answer: \$64

■ To find the sale price, calculate 20% of the original price of \$80 and subtract it from \$80. The key calculation step is: $\$80 * 0.20 = \16 , then $\$80 - \$16 = \$64$. The user's answer of \$65 is likely incorrect because they may have made a calculation error, such as incorrectly calculating the discount amount.

■ **Q6. A store buys a product for \$25 and sells it for \$30. What is the profit percentage?**

Your Answer: 12%

Correct Answer: 20%

■ To find the profit percentage, we need to calculate the profit made and then divide it by the cost price. The key calculation step is: Profit = Selling Price - Cost Price = $\$30 - \$25 = \$5$, and then Profit Percentage = $(\text{Profit} / \text{Cost Price}) * 100 = (\$5 / \$25) * 100 = 20\%$. The user's answer of 12% is likely incorrect because they may have made a calculation error, such as incorrect division or percentage calculation.

■ **Q7. A mixture of juice and water is in the ratio 3:5. If 3 cups of juice are added to the mixture, what is the new ratio of juice to water?**

Your Answer: 3:5

Correct Answer: 6:5

■ The correct answer is 6:5, which can be found by first determining the initial amount of juice and water. Let's assume the initial mixture has $3x$ cups of juice and $5x$ cups of water, and when 3 cups of juice are added, the new amount of juice is $3x + 3$. To find the new ratio, we set up a proportion, but since the amount of water remains the same ($5x$), we can simplify this by considering the ratio of juice to the constant amount of water. The user's likely mistake is assuming the ratio remains the same after adding more juice, which is incorrect because adding juice changes the proportion of juice to water. Key calculation step: Let's assume the initial amount of juice is $3x$ and water is $5x$. When 3 cups of juice are added, the new amount of juice is $3x + 3$, and we want this to be in the simplest ratio form.

■ **Q8. If it is true that all students are learners and some learners are teachers, what can be concluded?**

Your Answer: No students are teachers

Correct Answer: Some learners are students

■ The correct answer is "Some learners are students" because we know that all students are learners, which directly implies that some learners (at least the students) are indeed students. The user's answer, "No students are teachers", is incorrect because the given statements do not provide enough information to support this conclusion. The user likely made a mistake by misinterpreting the relationship between students, learners, and teachers, and incorrectly assumed that the statements implied a mutual exclusivity between students and teachers.

■ **Q9. If 2 men can do a job in 6 days, how many days will it take for 1 man to do the same job?**

Your Answer: 8

Correct Answer: 12

■ The correct answer is 12 because if 2 men can do the job in 6 days, it means that the total work required is $2 \text{ men} * 6 \text{ days} = 12 \text{ man-days}$. To find out how many days it will take for 1 man to do the job, we divide the total work by 1 man, which gives us $12 \text{ man-days} / 1 \text{ man} = 12 \text{ days}$. The user's likely mistake was not doubling the number of days when reducing the number of workers by half, which is a key step in this type of problem.

■ **Q10. A mixture of paint and water is in the ratio 3:4. If 3 cups of paint are added to the mixture, what is the new ratio of paint to water?**

Your Answer: 6:4

■ Correct! Well done.

■ MCQ SECTION (4/10 - 40.0%)

■ Q1. What is an interface in Java?

Your Answer:

A class that cannot be instantiated and cannot be inherited

Correct Answer:

A abstract class that cannot be instantiated

■ The correct answer is A) A abstract class that cannot be instantiated, because in Java, an interface is essentially an abstract class that cannot be instantiated on its own and is used to define a contract or a set of methods that must be implemented by any class that implements it. The user's answer is close, but not entirely accurate, as a class that cannot be instantiated and cannot be inherited doesn't fully capture the purpose of an interface, which is to be implemented by other classes. The user's mistake likely stems from a misunderstanding of the relationship between interfaces and inheritance, as interfaces can be implemented by multiple classes, but not inherited in the classical sense.

■ Q2. What is an example of a data type in Java?

Your Answer: For loop

Correct Answer: Int

■ The correct answer is "Int" because it is a fundamental data type in Java, used to represent integer values. The user's answer, "For loop", is incorrect because it is a control structure used for repetitive execution of a block of code, not a data type. The user likely mistook a programming concept for a data type, which are two distinct concepts in Java, with data types defining the type of value a variable can hold, such as Int, Double, or Boolean.

■ Q3. What is inheritance in Java?

Your Answer:

A subclass inherits properties and behavior from a parent class

■ Correct! Well done.

■ Q4. What are the data types in Java?

Your Answer: All of the above

■ Correct! Well done.

■ Q5. What is a RuntimeException in Java?

Your Answer:

A type of exception that is thrown at runtime

■ Well done!

■ Q6. What is an example of an exception in Java?

Your Answer: Creating an object

Correct Answer: Dividing a number by zero

■ The correct answer, "Dividing a number by zero", is an example of an exception in Java because it is an abnormal or unexpected event that occurs during the execution of a program, causing it to terminate abruptly. In Java, dividing a number by zero throws an `ArithmeticException`, which is a type of exception. The user's answer, "Creating an object", is likely incorrect because creating an object is a normal and expected operation in Java, whereas dividing by zero is an error that requires special handling.

■ Q7. What is a StackTrace in Java?

Your Answer:

A list of variables that were used before an exception was thrown

Correct Answer:

A list of methods that were called before an exception was thrown

■ The correct answer is A) A list of methods that were called before an exception was thrown. This is because a StackTrace in Java provides a record of the sequence of method calls that led to the exception, allowing developers to diagnose and debug the issue. The user's answer, C) A list of variables that were used before an exception was thrown, is incorrect because a StackTrace primarily focuses on the method call stack, not variable usage.

■ Q8. How can you compile and run a Java program?

Your Answer: Using a code editor or IDE

■ Well done!

■ Q9. What are common errors in Java?

Your Answer: Syntax errors

Correct Answer: All of the above

■ The correct answer is "All of the above" because Java, like any programming language, is prone to various types of errors, including syntax errors, logical errors, and runtime exceptions. Syntax errors occur when the code does not follow the language's rules, logical errors occur when the code does not produce the desired output, and runtime exceptions occur when an error occurs during the execution of the code. The user likely got it wrong by only considering one type of error, whereas the correct answer encompasses all possible common errors in Java.

■ Q10. What type of block is used to handle exceptions in Java?

Your Answer: Finally block

Correct Answer: All of the above

■ The correct answer is "All of the above" because in Java, exception handling is done using a combination of try, catch, and finally blocks. The try block contains the code that might throw an exception, the catch block handles the exception if it occurs, and the finally block contains code that is executed regardless of whether an exception occurred. The user's answer, "Finally block", is incorrect because it only accounts for one part of the exception handling process, not the entire process.

■ CODING SECTION (1.4/5 - 28.0%)

■ Q1. Write a Java program that defines a Shape class with methods to calculate area and perimeter, and a Circle class that inherits from Shape. **Input Fo...

Your Answer:

```
import java.util.Scanner;

class Shape {
    double side;

    Shape(double side) {
        this.side = side;
    }

    double area() {
        return side * side;
    }

    double perimeter() {
        return 4 * side;
    }
}

class Circle extends Shape {
    double radius;

    Circle(double radius) {
        super(radius);
        this.radius = radius;
    }
}
```

```

    }

    @Override
    double area() {
        return Math.PI * radius * radius;
    }

    @Override
    double perimeter() {
        return 2 * Math.PI * radius;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for (int i = 0; i < n; i++) {
            String type = sc.next();

            if (type.equals("Circle")) {
                double r = sc.nextDouble();
                Circle c = new Circle(r);
                System.out.println(c.area() + " " + c.perimeter());
            } else if (type.equals("Shape")) {
                double s = sc.nextDouble();
                Shape sh = new Shape(s);
                System.out.println(sh.area() + " " + sh.perimeter());
            }
        }

        sc.close();
    }
}

```

Correct Answer:

```

import java.util.Scanner;

class Shape {
    protected double radius;

    public Shape(double radius) {
        this.radius = radius;
    }

    public double calculateArea() {
        return Math.PI * radius * radius;
    }

    public double calculatePerimeter() {
        return 2 * Math.PI * radius;
    }
}

class Circle extends Shape {
    public Circle(double radius) {
        super(radius);
    }
}

public class Main {

```

```

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);
    int n = scanner.nextInt();
    scanner.nextLine();

    for (int i = 0; i < n; i++) {
        String shapeType = scanner.nextLine();
        double radius = scanner.nextDouble();
        scanner.nextLine();

        if (shapeType.equals("Circle")) {
            Circle circle = new Circle(radius);
            System.out.println(String.format("%.1f %.1f", circle.calculateArea(),
circle.calculatePerimeter()));
        } else if (shapeType.equals("Shape")) {
            Shape shape = new Shape(radius);
            System.out.println(String.format("%.1f %.1f", shape.calculateArea(),
shape.calculatePerimeter()));
        }
    }

    scanner.close();
}
}

```

■ The student's code failed to calculate the perimeter of the Circle shape correctly, as the actual output for the Circle's perimeter is 31.400000000000002, which is correct, but the Shape's perimeter is 40.0, whereas the expected output is 62.83185307179586. This suggests that the student's code did not override the perimeter method for the Shape class when it was not a Circle.

■ Q2. Write a Java program that reads a queue of integers from stdin and finds the maximum element.

****Input Format:** - Line 1: The number of elements in t...**

Your Answer:

```

import java.util.LinkedList;
import java.util.Queue;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        Queue<Integer> queue = new LinkedList<>();
        for (int i = 0; i < n; i++) {
            queue.add(sc.nextInt());
        }

        int max = Integer.MIN_VALUE;
        while (!queue.isEmpty()) {
            int val = queue.poll();
            if (val > max) {
                max = val;
            }
        }

        System.out.println(max);
        sc.close();
    }
}

```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over
        int max = Integer.MIN_VALUE;
        String[] queue = scanner.nextLine().split(" ");
        for (String num : queue) {
            int current = scanner.nextInt();
            scanner.nextLine(); // Consume newline left-over
            if (current > max) {
                max = current;
            }
        }
        System.out.println(max);
        scanner.close();
    }
}
```

■ The student's code failed 1 out of 5 test cases, indicating that it may not handle certain edge cases correctly. The provided code snippet appears to be mostly correct, but the issue might be related to the input or output handling. The test case failure suggests that the code may not produce the expected output for a specific input, but without more information, it's difficult to pinpoint the exact cause.

■ **Q3. Write a Java program that reads a string from stdin and handles any errors that occur during processing. ****Input Format:**** - Line 1: A string ****Outp...****

Your Answer:

```
import java.util.LinkedList;
import java.util.Queue;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        Queue<Integer> queue = new LinkedList<>();
        for (int i = 0; i < n; i++) {
            queue.add(sc.nextInt());
        }

        int max = Integer.MIN_VALUE;
        while (!queue.isEmpty()) {
            int val = queue.poll();
            if (val > max) {
                max = val;
            }
        }

        System.out.println(max);
        sc.close();
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        try (Scanner scanner = new Scanner(System.in)) {
```

```

        String input = scanner.nextLine();
        System.out.println(input);
    } catch (Exception e) {
        System.out.println("An error occurred: " + e.getMessage());
    }
}
}

```

■ The student's code failed because it's expecting an integer input (`int n = sc.nextInt();`) whereas the test case provides a string input ("hello"). This mismatch causes a runtime error. The code should be modified to read a string from stdin instead of an integer.

■ **Q4. Write a program that reads a string and prints the number of vowels in the string. ****Input Format:**** - Line 1: Input string (string) ****Output Format:**** ...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String line = sc.nextLine();

        int count = 0;
        for (char c : line.toLowerCase().toCharArray()) {
            if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') {
                count++;
            }
        }

        System.out.println(count);
        sc.close();
    }
}

```

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        int vowelCount = 0;

        for (char c : input.toLowerCase().toCharArray()) {
            if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') {
                vowelCount++;
            }
        }

        System.out.println(vowelCount);
        scanner.close();
    }
}

```

■ The student's code is written in Java, but the test environment is expecting Python code, as indicated by the error message referencing a Python file (`main.py`) and a `SyntaxError`. This mismatch is causing the code to fail. The code itself appears to be correct for counting vowels in a string, but it's not compatible with the test environment.

■ **Q5. Write a Java program that reads a string from stdin and validates it using a try-catch block. ****Input Format:**** - Line 1: A string ****Output Format:**** ...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        try {
            String input = sc.nextLine().trim();

            if (input.isEmpty()) {
                throw new IllegalArgumentException("String is empty");
            }
            if (!input.matches("[a-zA-Z]+")) {
                throw new IllegalArgumentException("String contains invalid characters");
            }

            System.out.println("Valid");

        } catch (IllegalArgumentException e) {
            System.out.println("Invalid");
        } finally {
            sc.close();
        }
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        try {
            String input = scanner.nextLine();
            System.out.println("Valid");
        } catch (Exception e) {
            System.out.println("Invalid");
        } finally {
            scanner.close();
        }
    }
}
```

■ The student's code failed 2 out of 5 test cases due to a runtime error, indicating an issue with the code's execution. The provided code snippet appears to be incomplete, but based on the test results, it's likely that the error occurs when trying to print the output, as the method name is misspelled as `System.out.prin`. This would cause a runtime error, resulting in the failed test cases.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.